

LIU44008 · Pragmatics II

Week 1

From Logic to Probability

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INTRODUCTION · WHERE WE ARE

WEEK 2

Game Theory

The strategic structure: agents, utilities, best response, equilibrium.

WEEK 3

Information Theory

The currency: surprisal, entropy, KL divergence, efficiency.

WEEKS 4 AND 5

Rational Speech Acts

The probabilistic implementation that composes the others.

WEEK 6

Cognitive Science

The resource bound, and the link from a model to data.

Diagnostic · Scalar implicature

Diagnostic · Presupposition

Thread · The semantics and pragmatics interface

The work of this term

WHAT YOU BUILD

You build the machinery: specify a phenomenon as a model, derive a prediction from it, and design the study that would test it.

ASSESSMENT, IN OUTLINE

Presentation 40%, in weeks 9–12. Essay 60%, 2,500 words, end of term. Deadline to follow.

Two sentences

Some of the students passed.

Alex stopped smoking.

Scalar implicature

The Horn scale { *some*, *all* } is ordered by **entailment**: *all* entails *some*.

A Gricean reasoner:

- | | |
|---------------------------|---|
| 1 · Observation | The speaker said <i>some</i> . |
| 2 · Premise | A cooperative speaker obeys Quantity. |
| 3 · So | She is not in a position to assert <i>all</i> . |
| 4 · Premise, quiet | She knows whether <i>all</i> holds. |
| 5 · Conclusion | Not all of the students passed. |

CANCELLABLE

Some of the students passed. In fact I have no idea how many. No contradiction.

Presupposition

RUSSELL 1905

The king of France is bald is **false**. The sentence asserts an existence claim, and that claim fails.

STRAWSON 1950

It is neither true nor false. The existence claim is a **precondition** on the sentence having a truth value at all.

Plain	<i>Alex stopped smoking.</i>
Negated	<i>Alex didn't stop smoking.</i>
Questioned	<i>Did Alex stop smoking?</i>
Embedded	<i>If Alex stopped smoking, he'll feel better.</i>

Deductive reasoning

Both specimens were built from logical machinery: an entailment scale on one side, a truth-value gap on the other. It is worth being explicit about what that machinery is.

AN ARGUMENT

Premises, a conclusion, and a rule licensing every step in between. Write $\Gamma \vdash A$ for: the conclusion A is derivable from the set of premises Γ .

VALIDITY

If every premise is true, the conclusion cannot be false.

A QUESTION

A question we might ask is: is natural language behaviour in context really like this? Is it all clean deductive logical inference?

A natural deduction system

Every connective gets a rule for building it in, and a rule for taking it out. Nothing else is permitted.

$\wedge$ introduction	From A and B, infer $A \wedge B$.
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$\wedge$ elimination	From $A \wedge B$, infer A. Or infer B.
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$\rightarrow$ elimination	From A and $A \rightarrow B$, infer B.
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$\rightarrow$ introduction	Assume A; derive B; discharge the assumption and conclude $A \rightarrow B$.
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$\neg$ introduction	Assume A; derive $\perp$; conclude $\neg A$.
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SESSION 1 · WORKED

A proof

1	$P \rightarrow Q$	<i>premise</i>
2	$Q \rightarrow R$	<i>premise</i>
3	P	<i>premise</i>
4	Q	<i>from 1, 3 by $\rightarrow E$</i>
5	R	<i>from 2, 4 by $\rightarrow E$</i>

FORMAL PROOF

Every line is a premise or is licensed by a named rule applied to earlier lines.

Monotonicity

If $\Gamma \vdash A$, then $\Gamma \cup \{B\} \vdash A$.

IN WORDS

Adding a premise never takes a conclusion away. Learn something new and everything you had already proved stays proved.

The derivation fails both tests

Put Grice's route from some to not all through the standard just set.

VALIDITY FAILS

Every premise can hold with the conclusion false. The speaker is cooperative, obeys Quantity, and says *some* because she has seen only two of the three students. All three passed. Premises true, conclusion false.

MONOTONICITY FAILS

Add one premise: *the speaker has no idea how many passed*. The conclusion disappears. No deductive system permits that.

GRICE

Grice gave two criteria for implicature. **Calculability**: the inference can be reconstructed as an argument. **Cancellability**: the reconstruction can be withdrawn.

Where the logical picture strains

GRADIENCE

Implicature rates vary continuously with context, and vary again across scales. A categorical rule predicts a rate of nought or one. The next slide shows what the data actually look like.

DEFEASIBILITY

Last hour's result. The first repair was attempted from inside logic: non-monotonic and default logics simply drop the rule that adding a premise preserves a conclusion. It is a real field, and it did not deliver a working pragmatics.

EPICYCLES

Step 4 on the derivation slide. It exists to make the argument go through, and it is stipulated rather than derived.

NOISE

Real signals are corrupted and comprehension is robust to it. A logic of the literal string has nothing to say about that.

Gradience, measured

There are ten circles. I tell you that some of the circles are white. How many are white?

WHAT PEOPLE ANSWER

Most say four or five. And two strikes people as less likely than six. Nobody has any prior knowledge about the colouring of ten circles, so the estimate is coming from the interpretation itself.

WHAT THE CATEGORICAL ACCOUNT PREDICTS

Some rules out none. The implicature rules out all. Everything in between is equally good. A flat line across the whole middle of the range.

WHAT THE RATINGS LOOK LIKE

Acceptability ratings for *Some of the As are Bs* rise, peak around a third to a half of the set, then fall away long before the top. Two independent experiments, the same shape (van Tiel; Degen and Tanenhaus).

AND SCALES DIFFER FROM EACH OTHER

some/all and *sometimes/always* yield the inference readily. *big/enormous* and *attractive/stunning* much less so. A theory built on entailment alone predicts no such difference (Doran et al.; van Tiel et al.).

The standard recipe

- 1 The speaker used *some*.
- 2 One reason she did not use *all*: she does not know whether it holds.
- 3 · **Competence Assumption** She knows whether it is *all* or *some but not all*.
- 4 So she knows the *all* sentence is false.
- 5 If she knows it, it is true.

WITHOUT STEP 3

You get only the weak inference: the speaker does not know that *all* is true. Something has to be stipulated to get the strong one.

THE ALTERNATIVE

Treat competence as a quantity the listener is uncertain about, and infers. Week 5 does exactly this, and the stipulation disappears.

Degrees of belief

THE SWAP

Truth values give way to probabilities. Derivation gives way to update. A hearer holds a distribution over what the speaker might mean, and an utterance reshapes it.

WHAT THAT BUYS

Gradient predictions for gradient data. Defeasibility for free, since new evidence shifts a posterior and nothing contradicts. Speaker uncertainty becomes representable instead of stipulated.

THE LOGIC IS CONTAINED, NOT DISCARDED

In week 5 you will prove a version of this: with flat priors, the probabilistic speaker prefers exactly the logically stronger message. The old picture is the special case.

NOT ONLY PRAGMATICS

The same turn happened in artificial intelligence after Pearl, in the psychology of reasoning after Oaksford and Chater, and in vision from Helmholtz onward.

What a model is

STATES

What the world could be. Three apples, of which some number are red.

ALTERNATIVES

What the speaker could have said. *all, some, none*. Change this set and the predictions change.

PRIORS AND UTILITIES

What was already likely, and what the speaker is trying to achieve. Weeks 4 and 2 respectively.

AND A PREDICTION THAT CAN BE WRONG

This is the part that matters. A model that accommodates any result predicts nothing. Every model you build this term has to say what would embarrass it.

Levels of explanation

Constraints	In a disjunction <i>A or B</i> , <i>A</i> may not entail <i>B</i> .	<i>what?</i>
Principles	Prefer the strongest reading of an ambiguous sentence.	<i>what?</i>
Maxims	Quantity: maximise the flow of relevant information.	<i>what and why?</i>
Reasons	Optimal language use: be rational, or try to be.	<i>why?</i>
Processes	Compute truth conditions, then apply pragmatics.	<i>how?</i>

THIS COURSE WORKS AT THE LEVEL OF REASONS

Derive the maxims and constraints from something more general rather than stipulating them. Anderson, on rational analysis: it "does not imply any actual logical deduction in choosing optimal behavior, only that the behavior will be optimized".

THE OTHER LEVELS STAY

A constraint can be correct as a description while a model at the level of reasons explains why it holds. Franke and Jäger treat these as different tools aimed at the same task. Processes, the bottom row, answer a separate question: week 6.

Your first model

THE TOOL

WebPPL, in the browser, through the probLang webbook. Nothing to install. You will modify supplied models rather than write them from scratch.

NOW, TOGETHER

Open the scalar implicature model. Run it. Read off what it predicts for *some*.

THEN MOVE ONE THING

Remove *all* from the alternatives and run it again. Predict what will happen before you press the button.

THE POINT OF THE EXERCISE

The inference lives in the alternative set. That is an idea you can now demonstrate rather than assert.

What gets built, and when

Week 2 · Game theory	Agents, utilities, best response. No probability yet.
Week 3 · Information theory	Surprisal and entropy. The currency the utilities are paid in.
Week 4 · Probability and priors	Bayes, common ground as a distribution, presupposition.
Week 5 · Rational Speech Acts	The three composed. Scalar implicature derived.
Week 6 · Cognitive science	The resource bound, and whether any of it costs time.
Week 8 · Models and data	Language models as instrument and as rival. Then your proposals.

Week 7 is reading week. Weeks 9 to 12 are your presentations.

The semantics and pragmatics interface

How much inference intrudes into truth-conditional content?

A THREAD, NEVER A WEEK

Flagged wherever a derivation forces the question. It gets no session of its own.

WHERE IT BITES

Embedded implicatures, presupposition projection, and the status of enriched content.

A MOVING BORDER

The boundary shifts as theories improve. Several phenomena once filed under pragmatics now have strong semantic analyses.

What you will be asked to produce

PRESENTATION · 40% · WEEKS 9-12

Propose the analysis. A phenomenon, its alternatives, its priors, its utilities, a prediction, and the rival account you are up against.

ESSAY · 60% · 2,500 WORDS

The same proposal written up, with the objections answered, plus the study that would discriminate your model from the rival, and a statement of what result would change your mind. End of term; deadline to follow.

WHY THEY ARE ONE OBJECT

You leave your presentation with a draft rather than a set of slides. Every question you are asked on the day is an objection you can answer in writing.

CHOOSING A TOPIC

A topic pool comes in week 5, so that reading week is usable. Week 8 is the full briefing.

WEEK 1

This week's readings

- Grice, H. P. (1975). Logic and Conversation. Revisited: the source of the inferential picture.
- Sperber, D. & Wilson, D. (1986/95). Relevance: Communication and Cognition , ch. 1. The cognitive reframing of Grice.
- Levinson, S. C. (1983). Pragmatics , introduction. A light recap of the shape of the field.

Primary-reading-led. Tight excerpting is expected for the harder primary texts.

Where this sits

THE SPECIMENS

Scalar implicature and presupposition, behaving differently under every test, and both born inside logic.

THE PROBLEM

Categorical machinery, gradient data, and a valid argument whose conclusion cancels.

THE MOVE

Degrees of belief, and a model that says what would embarrass it.

Next · Game theory

Lewis, signalling, and convention as equilibrium